MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 7 (4-8 Oct 2026) - Period 1	Unit 2 - 2.1 Creative Photo Editing (1/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 7 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Unit 1 exit quiz: security measures secure; ~30% unsure when an online image may be reused (IP). Today: quick 'Can I use this image?' check before editing.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students use GIMP selection tools (Rectangle, Ellipse, Free, Fuzzy, By Color, Intelligent Scissors, Foreground Select) to isolate a subject and change the background. CSTA 3A-DA-09; ISTE 6.b; Sprint 2.1 obj. 1.

Success criteria — *what will successful learning look / sound like?*

I can choose the right selection tool for an object's edges; I can isolate a subject with Foreground Select and place it on a new background with clean edges.

Key knowledge / skill / understanding:

Selection = the area edits apply to; tool choice depends on edges/colour; layers keep the original safe.

Likely misconception / access or challenge need:

'Delete the background' = erase by hand; EAL learners need tool names shown with icons.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Think-pair-share **Time:** 8 min

What students will do:

Students view 3 photos (sharp-edged box, single-colour sky, person with complex outline). Individually choose the best selection tool for each, agree with a partner, then share and justify.

Evidence tool: Mini-whiteboards

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Slide with 3 photos; mini-whiteboards; GIMP selection-tools icon card.

Support (low performers / SEN):

Icon card with tool name + picture + Arabic/English keyword.

Stretch (fast learners / gifted):

Explain when Intelligent Scissors beats Fuzzy Select and why.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Problem solving **Time:** 22 min

What students will do:

Master Challenge 1: open an image with several objects (unit image pack), use Foreground Select to isolate one object, then add a new background on a separate layer. Save as .xcf and export .png.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

GIMP 2.10; image pack go.qubitscs.com/PXl2; student book Sprint 2.1; QR video p7LO.

Support (low performers / SEN):

Step card with screenshots; start from a simple high-contrast image; teacher check-in at step 3.

Stretch (fast learners / gifted):

Use a hair/fur image and refine the mask; add a soft shadow so the subject sits naturally.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Cooperative learning **Time:** 7 min

What students will do:

Pairs swap screens and check each other's edge quality, layer use and file saved, using a 3-point checklist; give one 'glow' and one 'grow'.

Evidence tool: Peer-assessment checklist

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Printed / Teams 3-point peer checklist.

Support (low performers / SEN):

Checklist with sentence starters ('The edge is... because...').

Stretch (fast learners / gifted):

Suggest which other selection tool could improve the edge.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: show a before/after where a student photo of a school trip gets a new background. Share success criteria. Live-model on the board: Rectangle vs Fuzzy vs Foreground Select on one image; think aloud about edges; show the purple mask and Preview mask; key words on board: selection, mask, layer, foreground, export.

What is new or innovative in this lesson? *(ETEC looks for this)*

Students choose a background from a place that matters to them (home country, Makkah, AlUla, a city skyline) - their culture becomes the design brief; peer feedback in pairs via Teams.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Students painting the whole image instead of the subject; forgetting to click Select/Enter so nothing is selected; editing the original layer with no backup; saving only .xcf. Likely responses: 'Fuzzy select for everything'. Probe: 'What happens at a blurry edge?'

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket (Forms, 3 Qs): match tool to image situation; name the menu where Foreground Select lives; one thing still unclear. 2 min left for learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: upload exported .png + one sentence on which selection tool you used and why. Watch QR video 'Looks of image' (kCEN) for next lesson.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Think-pair-share got EAL students talking: the icon card helped them name tools. 24/28 (10/1) produced a clean cut-out; exit tickets showed tool choice understood.

What will I change next time?

Model Foreground Select twice: many skipped the 'click inside then paint' step. Pair a confident GIMP user with each new user from the start.

Students needing follow-up:

10/3 and 10/5: 4-5 students did not finish export (slow PC login). Recently arrived EAL students: check tool vocabulary next lesson.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy	Numeracy	Digital	Digital research	Real-life ①	Real-life ②	Inquiry	Thinking
read + write · 2-1-1-6	numbers / data · 2-1-1-6	students use it · 2-1-1-4	trusted sources · 2-1-1-4	application · 2-1-1-5	application · 2-1-1-5	experiment / research · 2-1-1-7	problem-solving · 2-1-1-7
☑1 ☐2 ☐3 ☑C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☐H

Create	Discussion	Pair / group	Learner choice	Self-assessment	Peer assessment	SEN / low	Gifted
something new · 2-1-1-7	expressing ideas · 2-1-1-8	cooperative work · 2-1-1-8	independent work · 3-2-1-6	2-2-1-1	2-2-1-1	specific support · 2-1-1-3	specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☑H	☑1 ☐2 ☑3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

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OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 7 (4-8 Oct 2026) - Period 2	Unit 2 - 2.1 Creative Photo Editing (2/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 7 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket P1: 85% matched selection tools; Foreground Select steps shaky for ~5 per class. 3-min re-demo, then filters and painting effects.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students change the look of images with GIMP filters: Old Photo (Defocus, Sepia, Mottle, Border), and painting effects GIMPressionist, Apply Canvas and Oilify. Sprint 2.1 obj. 2-4; Challenges 2-3; ISTE 6.b.

Success criteria — *what will successful learning look / sound like?*

I can turn a photo into an old-looking photograph with Filters > Decor > Old Photo; I can apply 2 artistic filters and explain what at least 2 settings change.

Key knowledge / skill / understanding:

Filters change pixel characteristics; Preview before OK; Work on Copy protects the original.

Likely misconception / access or challenge need:

Sepia = grayscale; more effects = better art. EAL: explain 'mottle', 'jagged', 'brush stroke'.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Discussion / dialogue **Time:** 6 min

What students will do:

Compare a modern and an old-look photo of the same monument. Pairs list 3 visible differences and guess which Old Photo options made them.

Evidence tool: Oral questioning

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Before/after monument slide; vocabulary list (border, jagged, filter).

Support (low performers / SEN):

Picture vocabulary; frame 'It looks ... because ...'.

Stretch (fast learners / gifted):

Predict what each Old Photo checkbox changes before testing.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Stations / carousel **Time:** 22 min

What students will do:

Challenge 3 + Challenge 2 as stations (7 min each): A) monument from your own heritage > Old Photo with border; B) GIMPressionist (Paper/Brush tabs); C) Apply Canvas + Oilify. Record one setting and its effect per station.

Evidence tool: Observation checklist

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

GIMP; image pack PXI2; Wikimedia Commons; QR videos kCEN, 1CA9; station cards.

Support (low performers / SEN):

Station cards with numbered screenshots; images pre-resized to 1200 px.

Stretch (fast learners / gifted):

Combine 2 effects on separate layers with opacity; add a credit line.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Gamification **Time:** 8 min

What students will do:

Quizizz 'Name that filter': see an edited image, pick the filter/setting used; bonus for the menu path.

Evidence tool: Digital quiz (Forms / Kahoot / Quizizz)

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Quizizz with 8 image questions.

Support (low performers / SEN):

Image-based answers; play in teams if needed.

Stretch (fast learners / gifted):

Bonus: exact menu path for each filter.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 7 min

Retrieval: 3-min re-demo of Foreground Select. Hook: 'How would this monument have looked in 1920?' Model Filters > Decor > Old Photo (Defocus, Sepia, Mottle, Border Size, Work on Copy), then Filters > Artistic > GIMPressionist with Preview on. Explain station rotation and the credit line: 'Photo: name, source'.

What is new or innovative in this lesson? *(ETEC looks for this)*

Culturally responsive brief: each student ages a monument from their own heritage, building a class 'world gallery'; stations keep everyone hands-on.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Watermarked images (IP link); filters applied to the wrong layer; large images freezing PCs on Oilify; no image credit. Likely: 'Sepia makes it black and white' / 'I see no change' (settings too small).

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Gallery: 3 students show work and name one setting. Exit check (Forms, 2 Qs) on Old Photo options + artistic filters. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: upload your old-photo monument + one painted image with source credits. Watch QR 'Enhance color' (aCT4).

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Heritage monuments got strong engagement: Egyptian, Pakistani, Filipino and Saudi students shared stories. Quizizz: 80% named the filter correctly.

What will I change next time?

Resize images before class (2 PCs froze on Oilify). Give the trusted image-site list at the start to save search time.

Students needing follow-up:

10/2 and 10/6: about 5 students did not finish station C. Two SEN students need printed station cards.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy	Numeracy	Digital	Digital research	Real-life ①	Real-life ②	Inquiry	Thinking
read + write · 2-1-1-6	numbers / data · 2-1-1-6	students use it · 2-1-1-4	trusted sources · 2-1-1-4	application · 2-1-1-5	application · 2-1-1-5	experiment / research · 2-1-1-7	problem-solving · 2-1-1-7
☑1 ☐2 ☐3 ☑C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H

Create	Discussion	Pair / group	Learner choice	Self-assessment	Peer assessment	SEN / low	Gifted
something new · 2-1-1-7	expressing ideas · 2-1-1-8	cooperative work · 2-1-1-8	independent work · 3-2-1-6	2-2-1-1	2-2-1-1	specific support · 2-1-1-3	specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☑C ☐H	☑1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 8 (11-15 Oct 2026) - Period 1	Unit 2 - 2.2 Advance Photo Editing

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 8 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Uploads: 80% submitted; credits improving. Exit check showed filters secure. Today: colour correction and repair tools, with an export routine.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students improve colour (Color Enhance, Color Balance), straighten an image, and repair photos with Red Eye Removal, Clone and Despeckle; export .png/.gif. Sprint 2.2 obj. 1-5; Challenges 1-3; CSTA 3A-DA-09.

Success criteria — *what will successful learning look / sound like?*

I can balance colour by range (shadows/midtones/highlights); I can straighten with the Measure tool; I can use 2 of 3 repair tools; I export in the right format.

Key knowledge / skill / understanding:

Colour = RGB values; Clone is a Paint tool copying pixels; Despeckle reduces noise; .gif limited to 256 colours.

Likely misconception / access or challenge need:

Brightness = colour balance; Clone = copy/paste a layer; red eye can't be fixed.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Storytelling **Time:** 6 min

What students will do:

'Tina's family photo' (exercise Q3): a tilted, dull flash photo with red eyes. Students list what is wrong and which tool fixes each problem on whiteboards.

Evidence tool: Mini-whiteboards

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Story slide; whiteboards.

Support (low performers / SEN):

Picture story with key words: flash, tilt, dull.

Stretch (fast learners / gifted):

Explain why red eye happens (light reflecting off the retina).

ACTIVITY 2 *student task + evidence of learning*

Strategy: Stations / carousel **Time:** 24 min

What students will do:

Four repair stations (6 min each): 1) Color Enhance + Color Balance, export .png; 2) Measure tool > Straighten; 3) Red Eye Removal with Threshold; 4) Clone a pattern (Ctrl+click) or Despeckle a noisy scan, export .gif.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

GIMP; images from pack; QR videos aCT4, hCYG, WCUC, KCIT, oCOW; station cards.

Support (low performers / SEN):

Station cards with screenshots; peer helper at each station; aim for 2 stations.

Stretch (fast learners / gifted):

Hide an unwanted object with Clone so no one can tell; compare .png vs .gif file sizes.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Cooperative learning **Time:** 7 min

What students will do:

Groups make a 'repair guide' card for one tool (menu path, 3 steps, one tip) and post it on Teams as a class help library for the Unit 2 assessment.

Evidence tool: Peer-assessment checklist

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Teams channel; card template.

Support (low performers / SEN):

Template with headings filled.

Stretch (fast learners / gifted):

Add a 'common mistake + fix' box.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: three damaged photos (dull, tilted, red eye). Model in 1 minute each: Colors > Color Balance by range; Measure tool > Straighten; select eyes > Filters > Enhance > Red Eye Removal; Ctrl+click clone source. Show the export routine: File > Export As > type > Export. Explain station timer.

What is new or innovative in this lesson? *(ETEC looks for this)*

Four-station repair lab plus a student-made help library on Teams that is reused for assessment revision.

Progress recorded / shared:

☑ Gradebook ☑ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Color Balance without choosing a range; measuring a vertical line; Red Eye without selecting eyes; Clone source not set; 'Save' vs 'Export' confusion; .gif losing colours. Likely: over-saturated colours - ask 'Does it look natural?'

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket (Forms): which menu holds Color Enhance; Clone belongs to which tool group; why did the .gif look different? Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: upload one before/after repaired photo. Complete Unit 2 'Match the following' and 'Fill in the blanks'.

5 REFLECT — After the lesson *Complete within a day. This is reflective-practice evidence and feeds your next Starting point.*

What worked — and how do I know?

Stations moved quickly with timers; repair cards used Arabic + English labels, which helped EAL peers. Most completed 3 of 4 stations.

What will I change next time?

Four stations in one period is tight: give a 1-minute reset and let slower groups do 2 stations. Explain the .gif palette limit before station 4.

Students needing follow-up:

10/1, 10/5: a few students confused Clone with copy/paste. Parents of 3 students informed about missing uploads.

COVERAGE CHECK *Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.*

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☑C ☐H	☐1 ☑2 ☐3 ☑C ☐H	☐1 ☑2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☐2 ☑3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 8 (11-15 Oct 2026) - Period 2	Unit 2 - 2.3 Advance Video Editing (1/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 8 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Repair stations: 3 of 4 done by most; .gif palette now understood. Today we move to video, starting with resolution and aspect ratio numbers.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students set a project profile in OpenShot (resolution, aspect ratio, frame rate), import media files and modify clips on the timeline (trim, split, arrange, transitions). Sprint 2.3 obj. 1-3; ISTE 6.a.

Success criteria — *what will successful learning look / sound like?*

I can pick a suitable profile and explain resolution, aspect ratio and frame rate; I can import media, trim/split clips and add a transition.

Key knowledge / skill / understanding:

Profile = resolution + aspect ratio + fps; timeline tracks layer media; transitions join clips.

Likely misconception / access or challenge need:

Higher fps always better; aspect ratio = resolution; editing the original files.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Discussion / dialogu **Time:** 8 min

What students will do:

Compare a 16:9 and a 9:16 clip of the same scene. Discuss: Which platform suits each? What do 1920x1080 and 30 fps mean? Calculate frames in a 10-second clip.

Evidence tool: Oral questioning

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Two sample clips; profile info slide.

Support (low performers / SEN):

Glossary card: resolution, aspect ratio, frame rate, with diagrams.

Stretch (fast learners / gifted):

Compare data size of 24 vs 60 fps for the same clip.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 20 min

What students will do:

Follow along: create a project with a chosen profile, File > Import Files, drag clips to tracks, trim and split with Razor, reorder, add 2 transitions between clips. Save project.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

OpenShot; sample videos go.qubitscs.com/vXP1; QR install 338u.

Support (low performers / SEN):

Clips pre-downloaded; step card; work with 2 clips only.

Stretch (fast learners / gifted):

Use 4 clips and match transition style to the story.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Role play **Time:** 8 min

What students will do:

Pairs: 'client and editor'. Client gives a 15-second brief (e.g., school sports day reel); editor explains which profile and transitions they would choose and why.

Evidence tool: Peer-assessment checklist

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Brief cards; checklist.

Support (low performers / SEN):

Brief cards with choices to pick from.

Stretch (fast learners / gifted):

Editor adds a justification using numbers (fps, resolution).

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: 15-sec school-event reel made in OpenShot. Model: Choose Profile; import; drag to track; Razor tool; drag-to-overlap transition; save .osp. Name the panels (Project Files, Timeline, Preview). Share success criteria.

What is new or innovative in this lesson? *(ETEC looks for this)*

Role play client/editor brings real-world media-industry language; numeracy through fps x seconds and resolution calculations.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Wrong profile mismatch leading to black bars; clips placed on the same track overlapping by accident; not saving project; OpenShot crash on slow PCs (save often). Response: 'the video looks stretched'.

Close / exit check *(summative + time for learners' questions)* Time: 8 min

Exit ticket: 1) define aspect ratio 2) frames in 12 s at 25 fps 3) one transition you used. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: plan a 20-second video (storyboard of 4 frames) on 'My culture in 20 seconds' - photos/clips must be your own or royalty-free.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Numbers made sense when calculated with the real clip; role play produced good vocabulary. Most saved a project with 2 transitions.

What will I change next time?

OpenShot was slow on 6 PCs - install check before the lesson and keep clips short (under 30 s). Give the profile glossary earlier.

Students needing follow-up:

Students with no OpenShot at home: provide lab time on Tuesday break. 10/3: 2 SEN students need 1:1 check on the timeline.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy	Numeracy	Digital	Digital research	Real-life ①	Real-life ②	Inquiry	Thinking
read + write · 2-1-1-6	numbers / data · 2-1-1-6	students use it · 2-1-1-4	trusted sources · 2-1-1-4	application · 2-1-1-5	application · 2-1-1-5	experiment / research · 2-1-1-7	problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☑C ☐H	☑1 ☐2 ☐3 ☑C ☐H	☐1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H

Create	Discussion	Pair / group	Learner choice	Self-assessment	Peer assessment	SEN / low	Gifted
something new · 2-1-1-7	expressing ideas · 2-1-1-8	cooperative work · 2-1-1-8	independent work · 3-2-1-6	2-2-1-1	2-2-1-1	specific support · 2-1-1-3	specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☑H	☑1 ☐2 ☑3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 9 (18-22 Oct 2026) - Period 1	Unit 2 - 2.3 Advance Video Editing (2/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 9 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Storyboards: 75% submitted; strong cultural ideas. Exit ticket fps calc 80% correct. Today: effects, audio, keyframe animation and export.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students add video effects, work with audio (echo, compressor), create keyframe animations and export in Simple vs Advanced mode. Sprint 2.3 obj. 3-6; Challenges 1-3; Mass. 9-12.CAS.c.4.

Success criteria — *what will successful learning look / sound like?*

I can add at least 2 video effects and 1 audio effect; I can animate a property with keyframes; I can export in Simple and Advanced mode and explain the difference.

Key knowledge / skill / understanding:

Keyframe = property value at a time; effects change clips; Advanced export controls codecs/bitrate.

Likely misconception / access or challenge need:

Animation needs drawing; Compressor makes the file smaller (it changes audio volume range).

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 10 min

What students will do:

Watch-and-do: right-click clip > Properties; move playhead; set Scale/Location; Insert Keyframe twice to create a zoom/slide animation.

Evidence tool: Mini-whiteboards

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

OpenShot; QR video 2CPA; sample clip.

Support (low performers / SEN):

Paused video steps; teacher at front with a 'helper desk'.

Stretch (fast learners / gifted):

Animate 3 properties at once (scale, rotation, alpha).

ACTIVITY 2 *student task + evidence of learning*

Strategy: Project-based task **Time:** 20 min

What students will do:

Build 'My culture in 20 seconds' from the storyboard: up to 5 animations, 2 video effects, one audio effect and a caption for accessibility. Export in Simple mode, then try Advanced.

Evidence tool: Rubric

Assessment: ☐ Diagnostic ☐ Formative ☑ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

OpenShot; student storyboards; royalty-free music list; rubric.

Support (low performers / SEN):

Shortened task: 3 clips, 2 animations, 1 effect; template project.

Stretch (fast learners / gifted):

Add closed captions and compare Simple vs Advanced export file sizes.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Discussion / dialogue **Time:** 8 min

What students will do:

Three volunteers present their 20-second video; class gives 'two stars and a wish' through Teams chat.

Evidence tool: Peer-assessment checklist

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Display board; Teams chat.

Support (low performers / SEN):

Sentence starters for feedback.

Stretch (fast learners / gifted):

Comment using technical terms (keyframe, codec, transition).

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 5 min

Hook: same clip with and without animation and audio effects. Model Effects tab drag-and-drop, audio Echo vs Compressor, Insert Keyframe, and File > Export Project > Export Video (Simple vs Advanced). Share rubric with 4 criteria.

What is new or innovative in this lesson? *(ETEC looks for this)*

Accessibility: every video must include a caption (Mass. 9-12.CAS.c.4); students celebrate their cultural backgrounds in their videos.

Progress recorded / shared:

☑ Gradebook ☑ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Keyframes not appearing because playhead not moved; music used without permission (copyright reminder); export taking too long - use 720p for class. Response: 'Compressor compresses the file'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit check (Forms): Simple vs Advanced export - give one difference; what does Echo do? Learners' questions + reminder of Unit 2 assessment next lesson.

Homework / platform *(e.g. Madrasati)*

Madrasati: upload the exported video (<= 50 MB). Revise Unit 2 using the class repair-guide cards and exercise questions.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Videos showed real pride: Sudanese, Indian, Filipino and Saudi students shared traditions respectfully. Captions done by 70%. Presentations built confidence.

What will I change next time?

Give export in 720p by default; 4 exports failed at 1080p. Spend more time on keyframe concept with a paper timeline first.

Students needing follow-up:

Students who did not export (approx. 3 per class) to finish at lunch lab. Remind 2 students about using copyrighted music.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy	Numeracy	Digital	Digital research	Real-life ①	Real-life ②	Inquiry	Thinking
read + write · 2-1-1-6	numbers / data · 2-1-1-6	students use it · 2-1-1-4	trusted sources · 2-1-1-4	application · 2-1-1-5	application · 2-1-1-5	experiment / research · 2-1-1-7	problem-solving · 2-1-1-7
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H

Create	Discussion	Pair / group	Learner choice	Self-assessment	Peer assessment	SEN / low	Gifted
something new · 2-1-1-7	expressing ideas · 2-1-1-8	cooperative work · 2-1-1-8	independent work · 3-2-1-6	2-2-1-1	2-2-1-1	specific support · 2-1-1-3	specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 9 (18-22 Oct 2026) - Period 2	Unit 2 - Assessment (Photo & Video Editing)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 9 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Videos submitted by ~85%; keyframes and export modes understood by most. Today's assessment checks all Unit 2 outcomes; extra time for SEN.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Assess Unit 2: selection tools, filters, colour tools, repair tools (GIMP) and OpenShot profiles, effects and export, through a written quiz and a practical task aligned to the ToS. Sprint 2.1-2.3.

Success criteria — *what will successful learning look / sound like?*

I answer MCQ, fill-in and matching items accurately; I complete a practical GIMP task to the given specification; I reflect honestly on my Unit 2 progress.

Key knowledge / skill / understanding:

All Unit 2 vocabulary: border, filters, jagged, pattern, brush stroke, compress, time-lapse, keyframe.

Likely misconception / access or challenge need:

Rushing the practical without reading the spec; EAL students: read questions aloud if needed.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Problem solving **Time:** 15 min

What students will do:

Written section (Forms): 6 MCQ, 4 fill-in-the-blanks, 5 match-the-following and 2 short answers (Simple vs Advanced export; how to make an image look old).

Evidence tool: Short quiz

Assessment: ☐ Diagnostic ☐ Formative ☑ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Forms quiz from Unit 2 exercise bank; ToS.

Support (low performers / SEN):

Read-aloud option; glossary allowed for EAL; +10% time for SEN.

Stretch (fast learners / gifted):

Extended answer: compare two background-removal methods.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Project-based task **Time:** 22 min

What students will do:

Practical: given an image pack, isolate a subject, place it on a new background, apply one colour correction and one artistic filter, then export as .png with a credit line.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☐ Formative ☑ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

GIMP; assessment image pack; practical spec + marking rubric.

Support (low performers / SEN):

Simplified spec with checklist ticks.

Stretch (fast learners / gifted):

Add a clone repair and a fuzzy border; justify choices in 2 lines.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Discussion / dialogue **Time:** 5 min

What students will do:

Self-assessment: students rate their confidence on each Unit 2 skill (traffic light) and write one target for the next unit.

Evidence tool: Self-assessment checklist

Assessment: ☐ Diagnostic ☐ Formative ☑ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Traffic-light self-assessment sheet (Forms).

Support (low performers / SEN):

Smiley-face scale + short phrases.

Stretch (fast learners / gifted):

Write a personal learning goal (ISTE 1.a).

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 4 min

Clear instructions: timing, rules for practical, where to save (Assessments folder, file name Class_Name_U2). Read through the rubric and success criteria. Remind students to check spelling of file names. No modelling; quick reassurance for anxious learners.

What is new or innovative in this lesson? *(ETEC looks for this)*

Assessment combines a digital auto-marked quiz with a practical rubric and student self-assessment, giving 3 types of evidence.

Progress recorded / shared:

☑ Gradebook ☑ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Students stuck on export; students opening the internet for answers; anxiety among new students - circulate calmly, read questions aloud on request, keep extra time for SEN plan.

Close / exit check *(summative + time for learners' questions)* Time: 4 min

Collect files; quick 1-minute check that every file is saved correctly. Learners' questions about next unit (Game Interface).

Homework / platform *(e.g. Madrasati)*

Madrasati: preview Unit 3 - install VS Code at home if possible (QR yCJk) and bring 1 idea for a simple web game.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Quiz average approx. 76%; practical rubric average level 3. Self-assessment matched teacher marks for most students - they know their gaps.

What will I change next time?

Practical spec was long for EAL learners: use bullet icons next time. Allow 5 more minutes for the practical.

Students needing follow-up:

Students below 50% (approx. 2-3 per class) for re-teach of selection tools; inform parents via gradebook comment.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☑2 ☐3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 10 (25-29 Oct 2026) - Period 1	Unit 3 - 3.1 Web Interface for the Game (1/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 10 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Unit 2 test avg 76%; selection tools weakest. Short re-teach sheet sent. New unit: start from students' game ideas from homework.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students set up the 'Bit by Bit' worm-game project: folder structure, naming (camel/kebab case), VS Code, and the HTML structure of the index page. Sprint 3.1 obj. 1; ISTE 4.a; Mass. 9-12.DTC.a.1; CSTA 3A-AP-16.

Success criteria — *what will successful learning look / sound like?*

I create the correct folder structure (html, css, js, images, audio); I name files in kebab case; I write a valid index.html with head, body, header, sections and form elements.

Key knowledge / skill / understanding:

HTML structures content; folders keep projects organised; naming conventions avoid broken links.

Likely misconception / access or challenge need:

Spaces in file names are fine; HTML styles the page; EAL: 'hierarchy', 'kebab case'.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Brainstorming **Time:** 7 min

What students will do:

Groups brainstorm what a 'catch the worm' vermicompost game needs on the start page (title, player name, glove choice, start button) and sketch the layout.

Evidence tool: Observation checklist

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

A3 paper/markers; vermicompost picture (real-life link).

Support (low performers / SEN):

Picture prompts of game screens.

Stretch (fast learners / gifted):

Add an accessibility feature to the sketch.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 22 min

What students will do:

Download the resource folder; create folders and kebab-case files; in VS Code write index.html structure (doctype, head, title, header, main, form, input, button) following the teacher.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

VS Code; resource folder go.qubitscs.com/cXAd; QR LCdB; source file h9pt.

Support (low performers / SEN):

Starter index.html with comments marking where to type.

Stretch (fast learners / gifted):

Add semantic tags and alt text; validate the page with the W3C validator.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Think-pair-share **Time:** 8 min

What students will do:

Pairs examine 3 file names and 3 code snippets; decide which follow camel/kebab case and which HTML has errors, then share fixes.

Evidence tool: Mini-whiteboards

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Slide with snippets; mini-whiteboards.

Support (low performers / SEN):

Highlighted lines to focus on.

Stretch (fast learners / gifted):

Rewrite a snippet using semantic HTML.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: play the finished worm game for 1 minute. Real-life link: vermicompost (earthworms recycle food waste - school garden). Model folder setup and VS Code basics (Explorer, Live Server/open in browser). Key words: camel case, kebab case, hierarchy, element, tag.

What is new or innovative in this lesson? *(ETEC looks for this)*

Students build a real browser game across Units 3-4, linked to sustainability (vermicompost) - a project that grows each lesson.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☐ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Files saved in Downloads not the project folder; missing closing tags; uppercase/space file names; students copying the source file without understanding. Response: 'my page is blank' - file not saved/opened.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: write the correct folder tree and one kebab-case file name; one question for the teacher. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: complete Challenge 1 page-structure sketch for a personal website (drawing + 5 HTML tags you would use).

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Game demo hooked everyone; vermicompost link sparked talk about recycling practices in different home countries. Folder structure done by 90%.

What will I change next time?

Check VS Code is installed on all lab PCs before the unit (lost 5 minutes in 10/4). Keep the starter file for SEN.

Students needing follow-up:

Students new to coding (mainly recent arrivals) need a 'tags cheat sheet'; 3 students did not save in the right folder.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☑1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 10 (25-29 Oct 2026) - Period 2	Unit 3 - 3.1 Web Interface for the Game (2/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 10 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket: 88% correct folder tree; closing-tag errors common. Today: start with a 'spot the bug' and build the game page.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students build the HTML structure of the game page (score, worm count, holes grid, tray, gloves, buttons) and plan a page structure from a layout. Sprint 3.1 obj. 1; Challenges 2-3; CSTA 3A-AP-17.

Success criteria — *what will successful learning look / sound like?*

I break a layout into header/container divs; I write the game page HTML with ids/classes for each part; my page opens without errors.

Key knowledge / skill / understanding:

div containers, id vs class, nesting creates hierarchy; plan structure before coding.

Likely misconception / access or challenge need:

id and class are the same; every element needs its own CSS file.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Problem solving **Time:** 8 min

What students will do:

Challenge 2: convert a given web page screenshot into a page-structure diagram (header, innerHeader1/2, container, innerContainer1/2).

Evidence tool: Open-ended task

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Printed screenshot; diagram template.

Support (low performers / SEN):

Template with boxes pre-drawn.

Stretch (fast learners / gifted):

Add class names using camel case.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 22 min

What students will do:

Code game.html: header with score and worm count, a holes container with 9 holes (div), worm tray, glove area, and buttons, each with ids/classes as in the source file.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

VS Code; QR nCfN; source file F9aL.

Support (low performers / SEN):

Partially completed game.html; colour-coded nesting guide.

Stretch (fast learners / gifted):

Use a loop-friendly naming pattern (hole1-hole9) and comment each section.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Jigsaw **Time:** 7 min

What students will do:

Expert groups each explain one part of the page (header, holes, tray, gloves) to a mixed group, using their code.

Evidence tool: Oral questioning

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Each student's code; jigsaw cards.

Support (low performers / SEN):

Expert role card with prompts.

Stretch (fast learners / gifted):

Explain how this structure will help JavaScript later.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Spot the bug: 3 HTML lines with missing closing tags. Model planning on paper first: draw boxes for the game page, then translate into nested divs. Explain id (unique) vs class (shared). Share success criteria.

What is new or innovative in this lesson? *(ETEC looks for this)*

Jigsaw: every student becomes an 'expert' explaining one part of the game structure - builds learner voice and collaboration across language groups.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Mis-nested divs; duplicate ids; students not linking files later; copying without understanding - ask 'what will this div hold?'. Likely response: 'Can I just use class for everything?'

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit check: label the structure diagram of the game page with correct tags; learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: Challenge 3 - draw a responsive image-gallery layout and write its HTML (no CSS yet).

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Planning on paper first reduced nesting errors a lot. Jigsaw groups mixing Arabic, Urdu and English speakers worked; experts were proud to explain.

What will I change next time?

Give more time for the game page; 30% needed the starter file. Shorten jigsaw to 2 rounds.

Students needing follow-up:

10/2 and 10/6: 4 students with duplicate ids to fix. Gifted coders could start the gallery challenge in class.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 11 (1-5 Nov 2026) - Period 1	Unit 3 - 3.2 Styling the Game Page

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 11 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Game pages complete for ~85%; gallery homework shows good layouts. Today: CSS Flexbox, then style the index and game pages in one period.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students use CSS Flexbox (display:flex, justify-content, align-items, gap, flex-wrap) to lay out and style the index page and the game page (glove area, holes, score). Sprint 3.2 obj. 1-4; CSTA 3A-AP-16, 3A-AP-21.

Success criteria — *what will successful learning look / sound like?*

I link style.css; I use at least 4 flex properties correctly; my glove area highlights the selected glove; my game page works at laptop and phone widths.

Key knowledge / skill / understanding:

Selectors target elements; Flexbox arranges items on a main and cross axis; classes reuse styles; flex-wrap helps responsive layout.

Likely misconception / access or challenge need:

justify-content and align-items are the same; flex goes on the child; fixed pixel widths are responsive.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*	ACTIVITY 2 *student task + evidence of learning*	ACTIVITY 3 *student task + evidence of learning*
Strategy: Gamification **Time:** 8 min	**Strategy:** Project-based task **Time:** 23 min	**Strategy:** Cooperative learning **Time:** 6 min
What students will do: Flexbox challenge: teams predict the flex rule that moves projected items into place, then test it live.	**What students will do:** Style the project: index page centred with flex; glove customisation area with a .selected state; game page holes grid, score board and tray using the project CSS guide.	**What students will do:** Pairs test each other's pages in DevTools device mode and record one layout issue + fix.
Evidence tool: Mini-whiteboards Assessment: ☑ Diagnostic ☐ Formative ☐ End Grouping: ☐ Individual ☐ Pair ☑ Group	**Evidence tool:** Rubric Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☑ Individual ☐ Pair ☐ Group	**Evidence tool:** Peer-assessment checklist Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☐ Individual ☑ Pair ☐ Group
Resources: Flexbox practice page; display board.	**Resources:** VS Code; QR 'Flex properties', 'Design the glove customization area'; project CSS sources.	**Resources:** Browser DevTools; checklist.
Support (low performers / SEN): Property card with arrows for the two axes.	**Support (low performers / SEN):** Starter CSS with blanks to fill (justify-content: ___).	**Support (low performers / SEN):** Checklist with pictures of good/bad layout.
Stretch (fast learners / gifted): Combine 3 properties in one answer.	**Stretch (fast learners / gifted):** Add a media query for under 600 px and CSS variables for a colour palette.	**Stretch (fast learners / gifted):** Suggest the exact CSS property to change.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: the same page without and with CSS. Link the stylesheet live. Draw main and cross axes on the board; model display:flex, justify-content, align-items, flex-wrap with instant refresh. Show the .selected class for gloves. Share rubric.

What is new or innovative in this lesson? *(ETEC looks for this)*

Students learn flex through a team game, then apply it straight to their own game, testing responsiveness with DevTools.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Stylesheet path wrong (no styling); flex applied to the child not the parent; images not showing (wrong path); spending all the time on colours. Likely: 'nothing moves'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: given a layout picture, write the 3 flex properties needed; name one responsive technique used. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: finish styling; upload screenshots at laptop and phone widths. Read student book on event listeners.

5 REFLECT — After the lesson *Complete within a day. This is reflective-practice evidence and feeds your next Starting point.*

What worked — and how do I know?

Gamified start worked; most applied flex correctly after the axis drawing. Several students used colours from their national flags thoughtfully.

What will I change next time?

Two sprints' styling in one period is tight: set a 12-minute checkpoint for the index page before moving to the game page.

Students needing follow-up:

Students with broken image paths (approx. 3 per class) and SEN students need a 1-page CSS reference.

COVERAGE CHECK *Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.*

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
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Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 11 (1-5 Nov 2026) - Period 2	Unit 4 - 4.1 Setting the Game Elements (1/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 11 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Styled pages uploaded by ~80%; image paths the main issue. Today JavaScript begins: event listeners to make the index page respond.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students script the index page with event listeners (click, input), use the DOM to read player name/glove choice and pass data to the game page with sessionStorage. Sprint 4.1 obj. 1-2; CSTA 3A-AP-16; Challenge 1.

Success criteria — *what will successful learning look / sound like?*

I link script.js; I add an event listener to a button; I read and store the player name and glove in sessionStorage and read it on the game page.

Key knowledge / skill / understanding:

DOM = page as a tree; events trigger functions; sessionStorage keeps data for the session.

Likely misconception / access or challenge need:

sessionStorage keeps data forever; onclick in HTML and addEventListener are unrelated.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 10 min

What students will do:

Watch-and-try: select an element with getElementById, add a click listener that changes text; console.log values in DevTools.

Evidence tool: Mini-whiteboards

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

VS Code; browser console; QR 'Event listeners'.

Support (low performers / SEN):

Code template with blanks; pair with a buddy.

Stretch (fast learners / gifted):

Use querySelectorAll to add listeners to all gloves.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Problem solving **Time:** 20 min

What students will do:

Script index page: read player name, store chosen glove, validate empty name, and on Start save to sessionStorage and open game.html; on game.html display the name.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Project source 'Communicate between the web pages'; student book 4.1.

Support (low performers / SEN):

Step-by-step flowchart and starter code.

Stretch (fast learners / gifted):

Challenge 1: 6x5 interactive grid highlighting clicked cell.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Think-pair-share **Time:** 7 min

What students will do:

Real-life link: where do websites remember you (cart, language choice)? Pairs compare sessionStorage vs cookies/localStorage and privacy concerns.

Evidence tool: Exit ticket

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Slide with examples; link to Unit 1 Data Privacy.

Support (low performers / SEN):

Picture examples of online shopping cart.

Stretch (fast learners / gifted):

Explain privacy risks of storing personal data.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: click 'Start' on the finished game - how does page 2 know my name? Model DOM tree on the board, event listener syntax, console.log debugging, sessionStorage.setItem/getItem. Key words: event, DOM, propagation, session.

What is new or innovative in this lesson? *(ETEC looks for this)*

Debugging routine with DevTools console taught explicitly; links JavaScript storage to data-privacy ethics from Unit 1.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☐ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Script tag in wrong place (element null); typos in ids; using = instead of ===; forgetting to check Console for errors. Likely: 'nothing happens when I click'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket (Forms): order 4 lines of an event listener; what sessionStorage does. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: finish index script; Challenge 1 grid (optional). Watch QR video 'Loops in JavaScript'.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Console debugging helped: students read errors instead of asking straight away. Privacy discussion drew on different cultural views of sharing data.

What will I change next time?

Model the script placement (end of body / defer) first - most 'null' errors came from that. More pair programming for beginners.

Students needing follow-up:

Approx. 4 per class still cannot pass data between pages; small-group help next lesson. Gifted: grid challenge done by 6 students.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
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Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 12 (8-12 Nov 2026) - Period 1	Unit 4 - 4.1 Setting the Game Elements (2/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 12 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket: 70% ordered listener correctly; sessionStorage understood by 80%. Today: loops and asynchronous timing for the game.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students use for/while loops and asynchronous methods (setTimeout, setInterval, clearInterval) to repeat actions and time events. Sprint 4.1 obj. 3-4; Challenges 2-3; CSTA 3A-AP-15.

Success criteria — *what will successful learning look / sound like?*

I write a for loop to repeat code; I use setTimeout/setInterval with milliseconds correctly; I stop a timer with clearInterval; my countdown works.

Key knowledge / skill / understanding:

Loops repeat; 1 s = 1000 ms; asynchronous code runs later, not in order.

Likely misconception / access or challenge need:

setTimeout pauses the program; loops always need a counter from 0.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*	ACTIVITY 2 *student task + evidence of learning*	ACTIVITY 3 *student task + evidence of learning*
Strategy: Inquiry / investigation **Time:** 8 min	**Strategy:** Problem solving **Time:** 20 min	**Strategy:** Cooperative learning **Time:** 8 min
What students will do: Predict-then-run: 4 short code snippets (loop, setTimeout order). Students predict console output, run, then explain differences.	**What students will do:** Challenge 2: countdown timer from 10 that disappears at 0 and shows a birthday message (or Eid/National Day greeting); then add a loop creating 9 holes dynamically.	**What students will do:** Pairs swap code and trace each other's timer with a trace table (time, value, output).
Evidence tool: Mini-whiteboards Assessment: ☑ Diagnostic ☐ Formative ☐ End Grouping: ☐ Individual ☑ Pair ☐ Group	**Evidence tool:** Practical / performance task Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☑ Individual ☐ Pair ☐ Group	**Evidence tool:** Peer-assessment checklist Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☐ Individual ☑ Pair ☐ Group
Resources: Snippets in browser console.	**Resources:** VS Code; QR 'Asynchronous methods'; source files.	**Resources:** Trace table template.
Support (low performers / SEN): Annotated snippets; buddy support.	**Support (low performers / SEN):** Countdown starter with comments; flowchart.	**Support (low performers / SEN):** Partially filled trace table.
Stretch (fast learners / gifted): Explain why setTimeout(...,0) still runs last.	**Stretch (fast learners / gifted):** Challenge 3: print primes up to n using loops and if-else.	**Stretch (fast learners / gifted):** Add an edge case (n = 0 or negative) to test.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 7 min

Hook: a countdown on the board - how do games keep time? Model a for loop with a trace table; model setInterval with a counter and clearInterval; convert seconds to ms on the board. Share success criteria.

What is new or innovative in this lesson? *(ETEC looks for this)*

Students choose the celebration message (birthday, Eid, Diwali, National Day) - culturally responsive context; trace tables build computational thinking.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Infinite loops freezing the tab; forgetting clearInterval; using seconds not ms; counter declared inside the interval. Response: 'my timer goes negative'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit check: write a setInterval countdown from 5 in pseudocode; convert 2.5 s to ms. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: finish countdown; try the primes challenge (optional). Review student book 4.2 'Start the game'.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Predict-then-run created real surprise and talk. Celebration choice made it personal and inclusive. 80% built a working countdown.

What will I change next time?

Show how to recover from an infinite loop (close tab) before practice. Trace tables need a worked example first.

Students needing follow-up:

Students who froze browsers repeatedly (10/4, 10/5) need loop-condition practice. Inform parents of 5 students behind on submissions.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☐2 ☐3 ☑C ☐H	☑1 ☑2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 12 (8-12 Nov 2026) - Period 2	Unit 4 - 4.2 Core Logic of the Game (1/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 12 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Countdowns working for ~80%; ms conversion now secure. Today: start the game, make worms pop out at random times in random holes.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students build core game logic: start the game, make worms pop out, set random times (Math.random range formula) and choose random holes. Sprint 4.2 obj. 1-4; CSTA 3A-AP-16; Activity 1.

Success criteria — *what will successful learning look / sound like?*

I start the game on a button click; I generate a random time between min and max; I choose a random hole; worms pop out repeatedly.

Key knowledge / skill / understanding:

Math.random() gives 0 to <1; min + rand*(max-min); Math.floor for whole numbers; array of holes.

Likely misconception / access or challenge need:

Math.random gives whole numbers; same worm can't appear twice in the same hole.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Problem solving **Time:** 8 min

What students will do:

Activity 1: calculate a random time between 2000 and 3000 ms for rand = 0.83 by hand, then verify in Chrome's Console.

Evidence tool: Mini-whiteboards

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Chrome Console; formula slide.

Support (low performers / SEN):

Calculator; worked example with 0.5.

Stretch (fast learners / gifted):

Write a function randomBetween(min,max) and test 5 values.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 22 min

What students will do:

Code along: startGame(), randomTime(), randomHole() and popOut() using setTimeout; worms appear in random holes until the game ends.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

VS Code; project source 'Core logic'; student book 4.2.

Support (low performers / SEN):

Code skeleton with functions named; pair programming.

Stretch (fast learners / gifted):

Prevent the same hole twice in a row (compare with lastHole).

ACTIVITY 3 *student task + evidence of learning*

Strategy: Debate **Time:** 7 min

What students will do:

Is a game fair if it is random? Teams argue for/against using real examples (dice, sports draws) and suggest how to make randomness feel fair.

Evidence tool: Oral questioning

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Debate prompt; examples.

Support (low performers / SEN):

Sentence starters.

Stretch (fast learners / gifted):

Link to probability: chance of the same hole twice with 9 holes.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: roll a dice - how can a computer 'roll'? Model Math.random, the range formula, Math.floor with an array of holes. Show the flow: start -> random hole -> pop out -> hide -> repeat. Share success criteria.

What is new or innovative in this lesson? *(ETEC looks for this)*

Numeracy and probability embedded in coding; students debate fairness of randomness, linking maths to game design ethics.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☐ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Using Math.round giving index 9 (out of range); setTimeout not called again; CSS class for 'up' worm missing. Likely: 'worms appear all at once'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: compute min + 0.25*(max-min) for 1000-4000; why Math.floor not Math.round? Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: finish pop-out logic; upload a 20-second screen recording or screenshot. Optional: Challenge 3 Start/Stop random colour.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Calculating by hand first made the code make sense; probability debate engaged quieter maths-strong students. 75% had worms popping by the end.

What will I change next time?

Show the off-by-one error deliberately next time. Give a skeleton to all students to save typing time.

Students needing follow-up:

Students whose worms don't hide (CSS class issue) - approx. 3 per class; peer mentors assigned for next lesson.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☐C ☐H	☑1 ☐2 ☐3 ☑C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 13 (15-19 Nov 2026) - Period 1	Unit 4 - 4.2 Core Logic of the Game (2/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 13 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket: 85% correct on range formula; Math.floor reason weaker. Today: scoring, tray, missed worms, results and edge cases.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students complete the game logic: score and worm count, add worms to the tray, calculate missed worms, display results with alert, return to index, use a gameOn variable and fix edge cases. Sprint 4.2 obj. 5-11; CSTA 3A-AP-21.

Success criteria — *what will successful learning look / sound like?*

My score increases only when a visible worm is clicked; missed worms = total - caught; results show at the end; clicks after game over do nothing.

Key knowledge / skill / understanding:

Variables store state; conditions check gameOn; edge cases are unusual inputs that break logic.

Likely misconception / access or challenge need:

Clicking fast counts multiple times is fine; testing once is enough.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 12 min

What students will do:

Code along: score and worm-count variables, click listener on worms, add worm image to tray, missed = total - caught.

Evidence tool: Mini-whiteboards

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Project source; student book 4.2.

Support (low performers / SEN):

Skeleton code; checklist of variables.

Stretch (fast learners / gifted):

Store high score in sessionStorage.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Problem solving **Time:** 18 min

What students will do:

Implement results alert, back to index, and gameOn flag; then hunt edge cases (double click, click after end, empty name) and fix at least two.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Edge-case card; browser console.

Support (low performers / SEN):

Edge-case card with hints for 2 cases.

Stretch (fast learners / gifted):

Write a short test plan table with 5 cases and expected results.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Cooperative learning **Time:** 7 min

What students will do:

Playtest swap: pairs play another pair's game and log bugs in a shared sheet (bug, steps, expected, actual).

Evidence tool: Peer-assessment checklist

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Shared Teams/Excel bug log.

Support (low performers / SEN):

Bug log with one example row.

Stretch (fast learners / gifted):

Prioritise bugs by severity.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: a buggy version where score goes up when clicking empty holes. Model variables for state, the gameOn check, and how a test plan works. Explain playtesting roles.

What is new or innovative in this lesson? *(ETEC looks for this)*

Professional playtesting with a shared bug log; students learn that software is tested by others before release.

Progress recorded / shared:

☑ Gradebook ☑ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Score updating on hidden worms; alert loops; gameOn never reset; students fixing bugs by deleting features. Response: 'it works on my computer' - test differently.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit check: write the formula for missed worms and one edge case you fixed. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: fix one bug from your playtest log; upload before/after code screenshot.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Bug log was a hit; students from different classes in later sections used the same format. 70% have a fully working game loop.

What will I change next time?

Cut the modelling to 10 minutes; give more time to edge cases. Prepare a 'common bugs' slide.

Students needing follow-up:

Students without a working game (approx. 5 per class): provide fixed source to continue; contact parents of 2 students with repeated missing work.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☐C ☐H	☑1 ☐2 ☐3 ☑C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☑1 ☐2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 13 (15-19 Nov 2026) - Period 2	Unit 4 - 4.3 Enhancements + 4.4 Animations

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 13 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Games with full logic for ~70%; bug logs good. Last lesson before the autumn break: polish the game with a custom cursor, popup, sound and animation.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students enhance the worm game: glove cursor, styled results popup, catch sound with mute, and a CSS @keyframes glove animation triggered from JavaScript. Sprint 4.3-4.4; CSTA 3A-AP-16, 3A-AP-19.

Success criteria — *what will successful learning look / sound like?*

My glove cursor works with a fallback; the results popup replaces alert(); a sound plays on catch and can be muted; the glove animates when catching a worm.

Key knowledge / skill / understanding:

cursor: url() with fallback; classList show/hide; Audio() plays after a click; @keyframes + add/remove class replays an animation.

Likely misconception / access or challenge need:

Any image size works as a cursor; music can autoplay; replaying an animation needs a page refresh.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Discussion / dialogue **Time:** 5 min

What students will do:

Compare a plain and an enhanced version of the game; agree 3 UX rules (clear, not annoying, player in control).

Evidence tool: Oral questioning

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Two demo builds.

Support (low performers / SEN):

Picture comparison with key words.

Stretch (fast learners / gifted):

Name an accessibility risk of sound-only feedback.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 26 min

What students will do:

Code along in 4 short steps: 1) glove cursor; 2) results popup with Restart; 3) catch sound + mute button; 4) @keyframes swing triggered by .catch class removed on animationend.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

VS Code; QR 'Customize the cursor', 'Create a customized popup', 'Audio elements', 'CSS animations'.

Support (low performers / SEN):

Popup, audio and keyframes CSS ready; students write the JS lines only.

Stretch (fast learners / gifted):

Popup scale-in animation and prefers-reduced-motion support.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Stations / carousel **Time:** 6 min

What students will do:

Quick playtest carousel: play two classmates' games and leave one 'star' and one 'wish' on a sticky note or Teams.

Evidence tool: Peer-assessment checklist

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Sticky notes or Teams post.

Support (low performers / SEN):

Feedback phrases card.

Stretch (fast learners / gifted):

Feedback names the exact feature and fix.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: play the plain game, then the enhanced one. Model each enhancement for about 1 minute: cursor CSS, popup classList toggle with z-index, new Audio().play() inside the click handler, @keyframes with 0/50/100% and animationend. Remind: save a backup first.

What is new or innovative in this lesson? *(ETEC looks for this)*

Four polish features in short modelled steps let every student finish a complete, playable game before the break; peers playtest it.

Progress recorded / shared:

☑ Gradebook ☑ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Cursor image too large; popup hidden (z-index); autoplay error; animation plays once only; changes breaking a working game. Likely: 'it worked before!' - restore the backup.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Summative reflection (Forms): what I learned in Units 2-4, my best feature, my next goal. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: upload the final zipped game + 3-line README. Over the break: share the game with family and note one suggestion.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Starters made four features possible in one period; 70% finished all four. Students chose sounds respectfully for classmates from different backgrounds.

What will I change next time?

Give enhancements and animations two periods next year; the playtest was too short.

Students needing follow-up:

Students who did not finish (2-4 per class): extension to the first week after the break; share exemplar games.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy	Numeracy	Digital	Digital research	Real-life ①	Real-life ②	Inquiry	Thinking
read + write · 2-1-1-6	numbers / data · 2-1-1-6	students use it · 2-1-1-4	trusted sources · 2-1-1-4	application · 2-1-1-5	application · 2-1-1-5	experiment / research · 2-1-1-7	problem-solving · 2-1-1-7
☑1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☑3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H

Create	Discussion	Pair / group	Learner choice	Self-assessment	Peer assessment	SEN / low	Gifted
something new · 2-1-1-7	expressing ideas · 2-1-1-8	cooperative work · 2-1-1-8	independent work · 3-2-1-6	2-2-1-1	2-2-1-1	specific support · 2-1-1-3	specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback